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Date of Experiment: 03/10/2024 & 10/10/2024

CM3191 Chemical Synthesis Experiments

Organic Experiment 1: Asymmetric Synthesis of Jacobsen Ligand from Racemic 1,2-diaminocyclohexane

1. Abstract

Jacobsen ligand, [(*R,R*)-*N,N'*-bis(3,5-di-*tert*-butylsalicylidene)-1,2-cyclohexanediamine], a well-known asymmetric catalyst for enantioselective epoxidation, was synthesized. Beginning with a racemic mixture of 1,2-diaminocyclohexane, resolution using L-(+)-tartaric acid and recrystallization were performed to obtain the (*R,R*)-1,2-diaminocyclohexane mono-(+)-tartrate salt diastereomer. The recrystalized product was then reacted with 3,5-di-*tert*-butylsalicylaldehyde to yield the enantiomerically pure Jacobsen ligand. The specific rotation was determined using polarimetry, and procedural improvements were proposed to increase the enantiomeric purity of the synthesized Jacobsen ligand.

2. Aim

The aim is to apply and practice techniques in asymmetric synthesis, starting from a racemic mixture of 1,2-diaminocyclohexane to produce the enantiomerically pure Jacobsen ligand. The differing physical property of diastereomers is leveraged for the chiral resolution of 1,2-diaminocyclohexane. This involves a reaction with L-(+)-tartaric acid, followed by recrystallization to isolate (*R,R*)-1,2-diaminocyclohexane mono-(+)-tartrate salt, a diastereomer of poorer water solubility. Moreover, the aim is to assess the purity of the synthesized Jacobsen ligand using ^{1}H NMR and to understand that polarimetry can measure optical purity, the critical parameter of asymmetric synthesis.

3. Introduction

Enantiomers are a pair of molecules that are non-superimposable mirror images of one another. Obtaining a specific enantiomer via asymmetric synthesis is vital since different enantiomers can have considerably different chemical and biological effects. This is particularly critical in the pharmaceutical industry as humans consist of chiral compounds like amino acids and carbohydrates which interact selectively to each enantiomer and produce varying physiological effects[1]. One enantiomer of a drug can be therapeutic whilst the other is lethal.

Utilizing asymmetric catalysts is a highly preferred method for asymmetric synthesis as incorporating a small quantity of chiral catalyst is sufficient to produce enantiomerically pure molecules. For instance, Jacobsen's catalyst is a frequently used asymmetric catalyst that allows enantioselective epoxidation of unfunctionalized olefins with enantiomeric excess (ee) higher than 90%[2]. The use of Jacobsen's catalyst significantly improved the success rate of enantiomerically pure epoxidation compared to the conventional 1,2-diaryldiamine-derived catalysts which only resulted ee value of 70%[2]. Jacobsen's catalyst consists of a tetradentate ligand that coordinates to manganese via two nitrogen and two oxygen atoms as well as a chlorine occupying the fifth coordination site[3]. Its chirality originates from the diaminocyclohexane backbone where C-N bonds attain a trans conformation, which in turn enables enantioselective epoxidation.

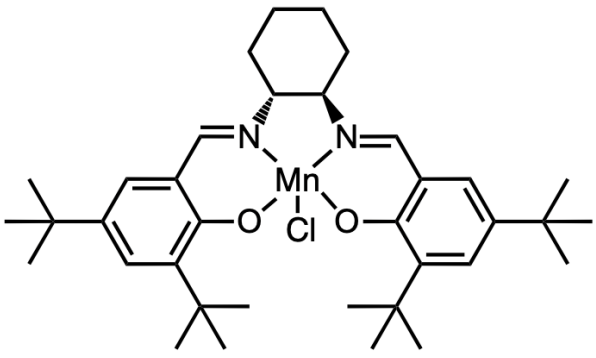


Figure 1: Structure of the Jacobsen's catalyst

The pivotal step for synthesizing the Jacobsen ligand is the resolution of racemic 1,2-diaminocyclohexane using enantiomerically pure L-(+)-tartaric acid to produce (*R,R*)-1,2-diaminocyclohexane mono-(+)-tartrate salt. The isolation of pure diastereomeric tartrate salt can be done by single-solvent recrystallization using water as the solvent.

Unlike enantiomers that have identical chemical and physical properties except for optical activity, diastereomers exhibit different physical properties. Varying water solubility of diastereomers can be employed for 1,2-diaminocyclohexane resolution. The tartrate salt consisting of (*R,R*)-1,2-diaminocyclohexane is poorly soluble in water while (*S,S*)-1,2-diaminocyclohexane is highly soluble[4]. Hence, after forming a saturated solution using boiling water, (*R,R*)-1,2-diaminocyclohexane mono-(+)-tartrate salt can be precipitated selectively by cooling down the solution. This is because of its poorer solubility and the Le Chatelier's Principle claiming that solubility decrease is accompanied when temperature falls as the system releases heat energy by causing solute precipitation[5].

Following the resolution of the tartrate salt and the reaction with 3,5-di-*tert*-butylsalicylaldehyde, enantiomeric purity of the resultant Jacobsen ligand can be assessed using polarimetry. Polarimeter utilizes linearly polarized monochromatic light which travels through the sample and determines how much the detector should rotate to allow maximum light transmission[6]. The observed rotation, α, is related to the specific rotation, $[\alpha]$, as given to the following expression[4] where c is the concentration in g/mL and l is the cell length in dm:

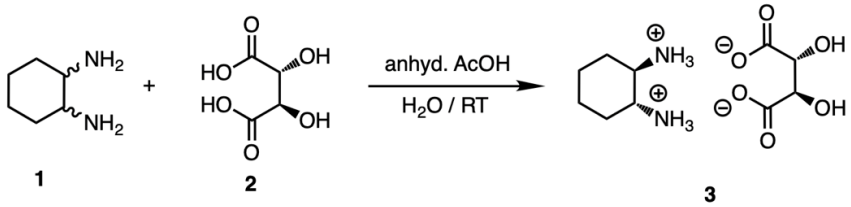
$$[\alpha] = \frac{\alpha}{cl}$$

Subsequently, the experimentally determined $[\alpha]$ can be compared with the theoretical value to determine the ee which measures enantiomeric purity and the ratio of enantiomers present:

$$\%\, ee = \frac{\left|[\alpha]_{\text{experimental}}\right|}{\left|[\alpha]_{\text{theoretical}}\right|} \times 100\% = \frac{|R-S|}{R+S} \times 100\%$$

4. Procedure

4.1 Resolution of racemic 1,2-diaminocyclohexane



1 2 3

Scheme 1: Resolution of 1,2-diaminocyclohexane

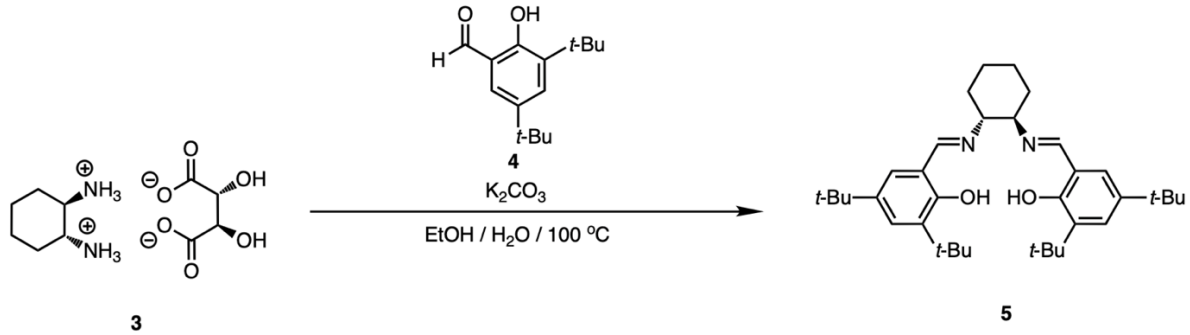
In a 100 mL round bottom flask containing a magnetic stirrer, 3.75 g (25 mmol, 1.0 eq) of L-(+)-tartaric acid **2** was added and dissolved in 15 mL of deionised water. Just above the heating plate, the round bottom flask was clamped and the rotating speed was set to 400 rpm to begin stirring. 6.1 mL (49.7 mmol, 2.0 eq) of racemic 1,2-diaminocyclohexane **1** was added dropwise using a dropper to result in an opaque yellow-orange solution. At room temperature, the reaction mixture was stirred for 5 minutes until it became a yellow transparent solution, indicating complete dissolution. The formation of water droplets inside the round bottom flask surface also suggested the progress of dissolution, an exothermic process.

After complete dissolution, 2.5 mL (42.7 mmol, 1.7 eq) of anhydrous acetic acid was added. Precipitation was observed with the reaction mixture transitioning to an opaque yellow slurry. The reaction mixture was cooled in an ice water bath for 30 minutes and was left undisturbed to allow crystal growth. The product was then separated by suction filtration. A metal spatula was used to transfer the product in the round bottom flask to the Büchner funnel. The solid product was washed with 5.0 mL of cold deionised water and 5 × 5 mL of room temperature methanol until the product appeared white without any yellow particles.

Recrystallization with boiling water was performed to isolate *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt **3**. The product in the Büchner funnel was transferred to a 50 mL conical flask and the mass of the crude product was measured. To the conical flask, 10 mL of deionised water was initially added to prevent thermal decomposition of the solid product and 3 pieces of boiling chips were also introduced to avoid bumping. Along with the conical flask, a 50

mL beaker containing 25 mL of deionised water was placed on the heating plate. As the mixture in the conical flask boiled, small amount of boiling deionised water was transferred using a dropper to the conical flask and swirled to allow dissolution. This step was repeated until just enough water was added to completely dissolve the crude product, resulting in a saturated solution. The conical flask was cooled in an ice water bath for 15 minutes until white crystals formed. The crystals were collected by suction filtration. To the filtrate, 15 mL of methanol was added to reduce the product solubility and was filtered again to obtain more *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt **3**.

4.2 Synthesis of Jacobsen ligand



Scheme 2: Synthesis of Jacobsen ligand

Into a 100 mL round bottom flask containing a magnetic stirrer, 0.6753 g (2.548 mmol, 1.0 eq) of (*R,R*)-1,2-diaminocyclohexane mono-(+)-tartrate salt **3** was added along with 0.70 g (5.096 mmol, 2.0 eq) of K_2CO_3 and 6 mL of deionised water. The mixture was stirred until full dissolution was achieved to result in a transparent solution, and then 22 mL of ethanol was added. The round bottom flask was placed into an oil bath and connected with a condenser. The reflux was started at 150 °C. In a 50 mL beaker, 1.21 g (5.096 mmol, 2.0 eq) of 3,5-di-*tert*-butylsalicylaldehyde was added and dissolved in 10 mL ethanol. This aldehyde solution was heated and then added to the round bottom flask using a dropper. 2 × 2.0 mL of hot ethanol was used for rinsing and transferred into the round bottom flask. The mixture was refluxed for 1 hour with mixing at high speed of 1000 rpm since the mixture consisted of chunks of solid.

After the reflux, 6.0 mL of deionised water was added and the reaction mixture was cooled in an ice water bath for 30 minutes. Suction filtration was performed to collect the yellow solid product which was washed with 4 × 5.0 mL of ethanol. The product was transferred to a separatory funnel, dissolved in 25 mL of CH_2Cl_2 and extracted with 2 × 5.0 mL of deionised water. The separated organic layer was washed with 5 mL of saturated NaCl solution and dried using Na_2SO_4. Na_2SO_4 was removed by simple gravity filtration and the solvent was removed using the rotary evaporator. A yellow solid, Jacobsen ligand [(*R,R*)-*N,N'*-bis(3,5-di-*tert*-butylsalicylidene)-1,2-cyclohexanediamine] **5** was obtained.

4.3 ^{1}H NMR and polarimetry

10 mg of the product was weighed and added to the NMR tube along with 0.6 mL of $CHCl_3$ for ^{1}H NMR. The optical purity of Jacobsen ligand was measured via polarimetry. 0.5537 g of Jacobsen ligand was measured and dissolved in CH_2Cl_2 in a 10 mL volumetric flask. The 1 dm cell was washed twice with the Jacobsen ligand solution. Then the cell was filled up to its neck with the solution, ensuring there were no air bubbles. The optical activity was measured using the polarimeter and the result was recorded. After the measurement, the solution was disposed and the cell was washed with CH_2Cl_2.

5. Results and Discussion

5.1 Percentage yield

5.1.1 Percentage yield of *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt

Compared to the literature yield of 52%[7], the experimental yield is considerably lower with only being 20.6% (calculation in Section 8.1.1). The deviation could be attributed to two main reasons. Firstly, large volume of water was used to assist the crude product transfer for suction filtration. Because the tartrate salt readily dissolves in water, the portion dissolved in water would have been filtered through and lost. The loss can be minimized by only using a metal spatula to break and transfer the solid product. For rinsing, a small volume of ice-cold water may be used as the solubility is decreases at lower temperatures or methanol can be used in which tartrate salt has poor solubility.

In addition, the solution resulting from recrystallization may have been unsaturated. Only 0.5022 g of crystals were obtained which is significantly less than 0.8917 g - half the crude product mass that should have been resulted assuming 50:50 ratio for isomers of trans 1,2-diaminocyclohexane. Obtaining a saturated solution is essential as it allows the maximum quantity of solute to be precipitated and crystallized. Subsequent addition of methanol to the initial filtrate decreased the solubility and white opaque solid appeared (0.1731 g), suggesting that excessive amount of water was indeed added during recrystallization. This observation emphasizes the importance of adding only a small amount of boiling water each time to acquire a saturated solution. Such practice is also important because adding large volumes of methanol can lead to the precipitation of water-soluble impurities and the other diastereomer, *(S,S)*-1,2-diaminocyclohexane mono-(+)-tartrate salt, reducing the optical purity as a result. This hypothesis is supported by visually inspecting the second filtration product that appeared as an opaque powder unlike that of the first filtration where crystals were large and transparent.



Figure 2: Image of the product from the initial filtration (left) and the second filtration (right)

5.1.2 Percentage yield of Jacobsen ligand

In contrast to the 1,2-diaminocyclohexane resolution, the yield for Jacobsen ligand synthesis is high being 96.0% (calculation in Section 8.1.2). More importantly, the yield is identical to the theoretical yield of 96%[2], implying that the reaction was done in a controlled manner without significant malpractice. Yet, ^{1}H NMR analysis suggests that the percentage yield is somewhat overestimated due to the presence of solvent peaks as well as those corresponding to the Jacobsen ligand (detailed discussion in Section 5.2).

5.2 ^{1}H NMR analysis

Chemical shift (ppm)	Multiplicity	Coupling constant (Hz)	Integration	Proton assignment
13.72	s	-	1.82	H_a
8.33	s	-	2.00	H_b
7.33	d	2.5	2.00	H_c
7.01	d	2.5	2.00	H_d
3.40-3.29	m	-	2.01	H_e
1.99-1.89	m	-	4.15	H_f
1.81-1.70	m	-	2.10	H_g
1.53-1.47	m	-	1.83	H_g
1.43	s	-	18.32	H_h
1.26	s	-	18.18	H_i

Table 1: ^{1}H NMR peak assignment for protons of the Jacobsen ligand

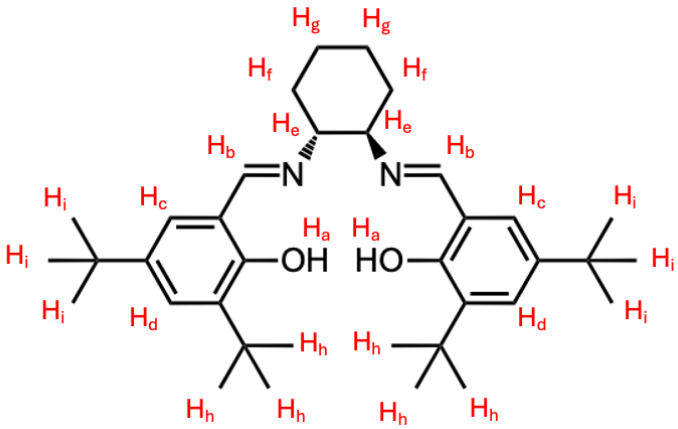


Figure 3: Structure of the Jacobsen ligand with the assignment of protons

All the characteristic peaks of Jacobsen ligand were identified and chemical shift and integration values matched that of the reported ^{1}H NMR results[8]. The only difference was the reporting of cyclohexane protons. Instead of reporting H_g separately, the literature reported combined integration value of six for 2.0-1.4 ppm range (H_f & H_g) and twenty for 1.45 ppm (H_g & H_h)[8]. In the experimental results, H_g peaks appeared at 1.81-1.70 and 1.53-1.47 ppm ranges of which the integration value summed up to be 3.93, matching the number of H_g protons. The non-equivalent chemical shifts result from the carbons that are chiral centres bonded to nitrogen. The trans conformation of nitrogens results in a slightly different shielding effect, giving two set of peaks for H_g protons.

Alongside peaks of the Jacobsen ligand, solvent peaks were observed. The peak at 5.32 and 1.59 ppm belongs to CH_2Cl_2 and H_2O, respectively. Moreover, a singlet at 1.37 ppm and a triplet at 1.29 ppm implies the presence of ethanol. Excluding CH_2Cl_2 existing as a trace amount, the presence of solvent in the final product can be calculated. Based on the integration values and the number of protons in each solvent molecule, it was determined that the molar fraction of H_2O and ethanol relative to Jacobsen ligand are approximately 0.50 and 0.24, respectively. Though the molecular weight of the Jacobsen ligand is significantly greater, it should be noted that the high molar fraction of solvent existing in the final product implies that the percentage yield is slightly lower than 96.0%.

5.3 Optical activity

Optical activity, the ability to rotate plane-polarized light, was measured using the polarimeter. The specific rotation was calculated to be -314.2° at 589 nm which resulted in an enantiomeric excess (ee) value of 99.75% (calculation in Section 8.3). The extremely high ee implies that optically pure [(*R,R*)-*N,N'*-bis(3,5-di-*tert*-butylsalicylidene)-1,2-cyclohexanediamine] was obtained from the asymmetric synthesis. The reason why ee did not reach 100% could be due to trace amount of *(S,S)*-1,2-diaminocyclohexane mono-(+)-tartrate salt resulting from recrystallization (Section 5.1.1). Another possibility is the solvent interaction such as hydrogen bonding and electron donation that cause optical rotation to deviate from the theoretical value despite having optically pure product[9]. As the ^{1}H NMR spectrum indicated the presence of solvents such as water and ethanol, solvent interaction is also a valid reasoning for ee being less than 100%.

6. Conclusion

Jacobsen ligand, [(*R,R*)-*N,N'*-bis(3,5-di-*tert*-butylsalicylidene)-1,2-cyclohexanediamine], was successfully obtained through the resolution of racemic 1,2-diaminocyclohexane and subsequent reaction with 3,5-di-*tert*-butylsalicylaldehyde. The ^{1}H NMR analysis proved that the synthesized Jacobsen ligand is pure and no impurity is present though residual solvent was identified. Moreover, the polarimetry result indicated that the product is also optically pure with 99.8% ee. Still, when performing recrystallization, greater caution is needed in order to acquire a saturated solution and subsequently increase the yield of 1,2-diaminocyclohexane resolution step.

7. References

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8. Appendices

8.1 Percentage yield calculation

8.1.1 Yield of *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt

Balanced chemical equation for 1,2-diaminocyclohexane resolution:

$C_6H_{10}(NH_2)_2 + C_4H_6O_6 \rightarrow C_6H_{10}(NH_2)_2 \cdot C_4H_6O_6$

Moles of *(R,R)*-1,2-diaminocyclohexane used:

6.1 (mL) × 0.931 (g/mL) ÷ 114.192 (g/mol) = 49.733 mmol

Since 1,2-diaminocyclohexane used is a racemic mixture, using 50:50 for cis:trans ratio, the number of moles of trans 1,2-diaminocyclohexane is 24.866 mmol. Noting that 25 mmol of L-(+)-tartaric acid was used, trans 1,2-diaminocyclohexane is the limiting reagent.

Mass of crude 1,2-diaminocyclohexane mono-(+)-tartrate salt: 1.7834 g

Moles of crude 1,2-diaminocyclohexane mono-(+)-tartrate salt produced:

1.7834 (g) ÷ 264.28 (g/mol) = 6.7481 mmol (5 s.f.)

Percentage yield of crude 1,2-diaminocyclohexane mono-(+)-tartrate salt:

6.7481 (mmol) ÷ 24.866 (mmol) × 100% = 27.138% (5 s.f.)

Mass of pure *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt: 0.6753 g

Moles of pure *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt produced:

0.6753 (g) ÷ 264.28 (g/mol) = 2.5552 mmol (5 s.f.)

Percentage yield of pure *(R,R)*-1,2-diaminocyclohexane mono-(+)-tartrate salt:

2.5552 (mmol) ÷ 12.433 (mmol) × 100% = 20.552% (5 s.f.)

8.1.2 Yield of Jacobsen ligand

Balanced chemical equation for Jacobsen ligand synthesis:

$C_6H_{10}(NH_2)_2 \cdot C_4H_6O_6 + 2HOC_6H_2[C(CH_3)_3]_2CHO$ ➔ $[[(CH_3)_3C]_2C_6H_2\text{-}2\text{-}(OH)CH{=}N]_2C_6H_{10}$

Mass of Jacobsen ligand produced: 1.3381 g

Moles of Jacobsen ligand produced: 1.3381 (g) ÷ 546.83 (g/mol) = 2.4470 mmol (5 s.f.)

As 5.171 mmol of 3,5-di-*tert*-butylsalicylaldehyde was used, (*R,R*)-1,2-diaminocyclohexane mono-(+)-tartrate salt is the limiting reagent.

Percentage yield of Jacobsen's ligand: 2.4470 (mmol) ÷ 2.5484 (mmol) × 100% = 96.021% (5 s.f.)

8.2 ^{1}H NMR

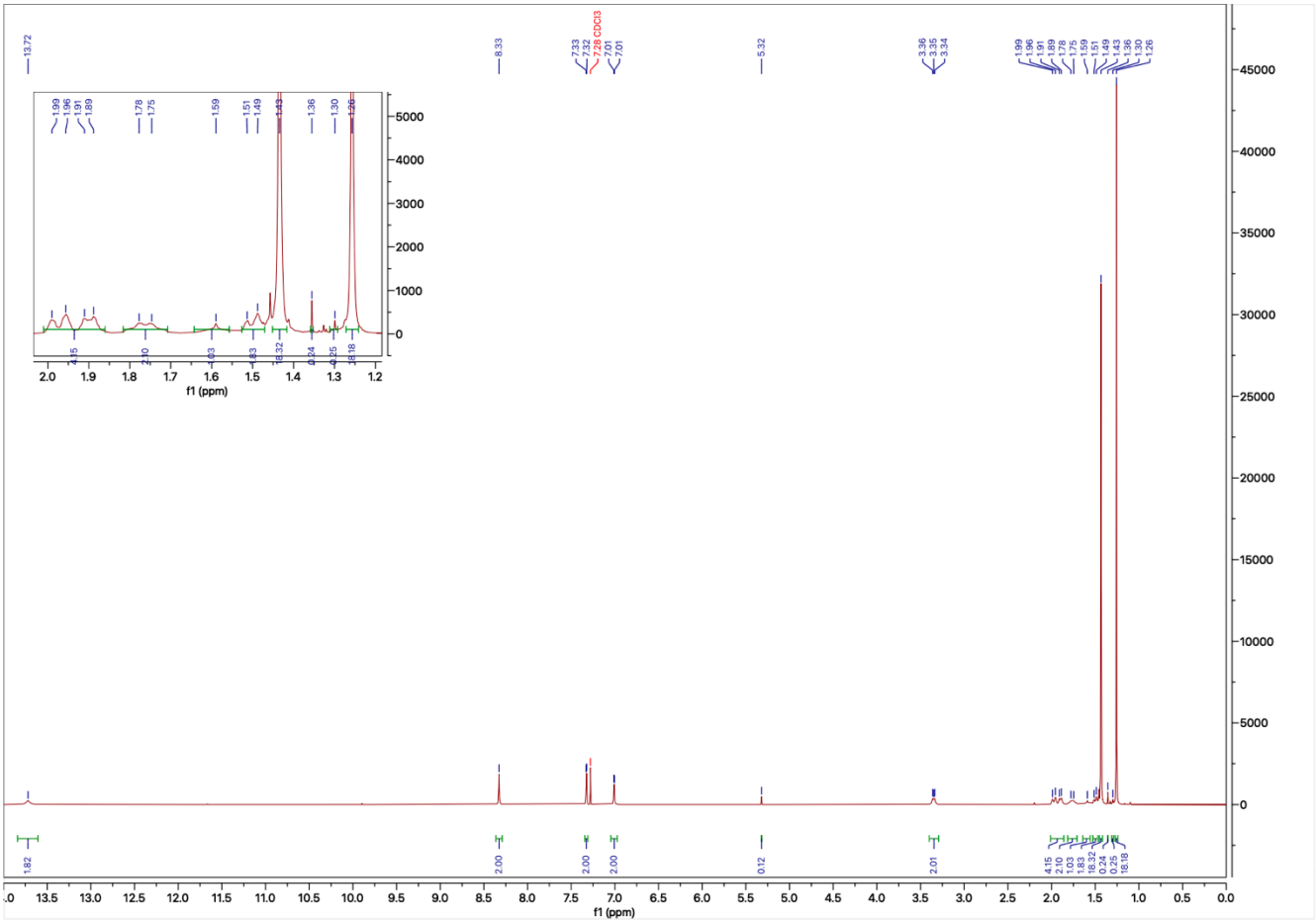


Figure A1: 1H NMR spectrum of the Jacobsen ligand

8.3 Optical activity

Mass of Jacobsen ligand used for polarimetry: 0.5537 g

Concentration of Jacobsen ligand solution:

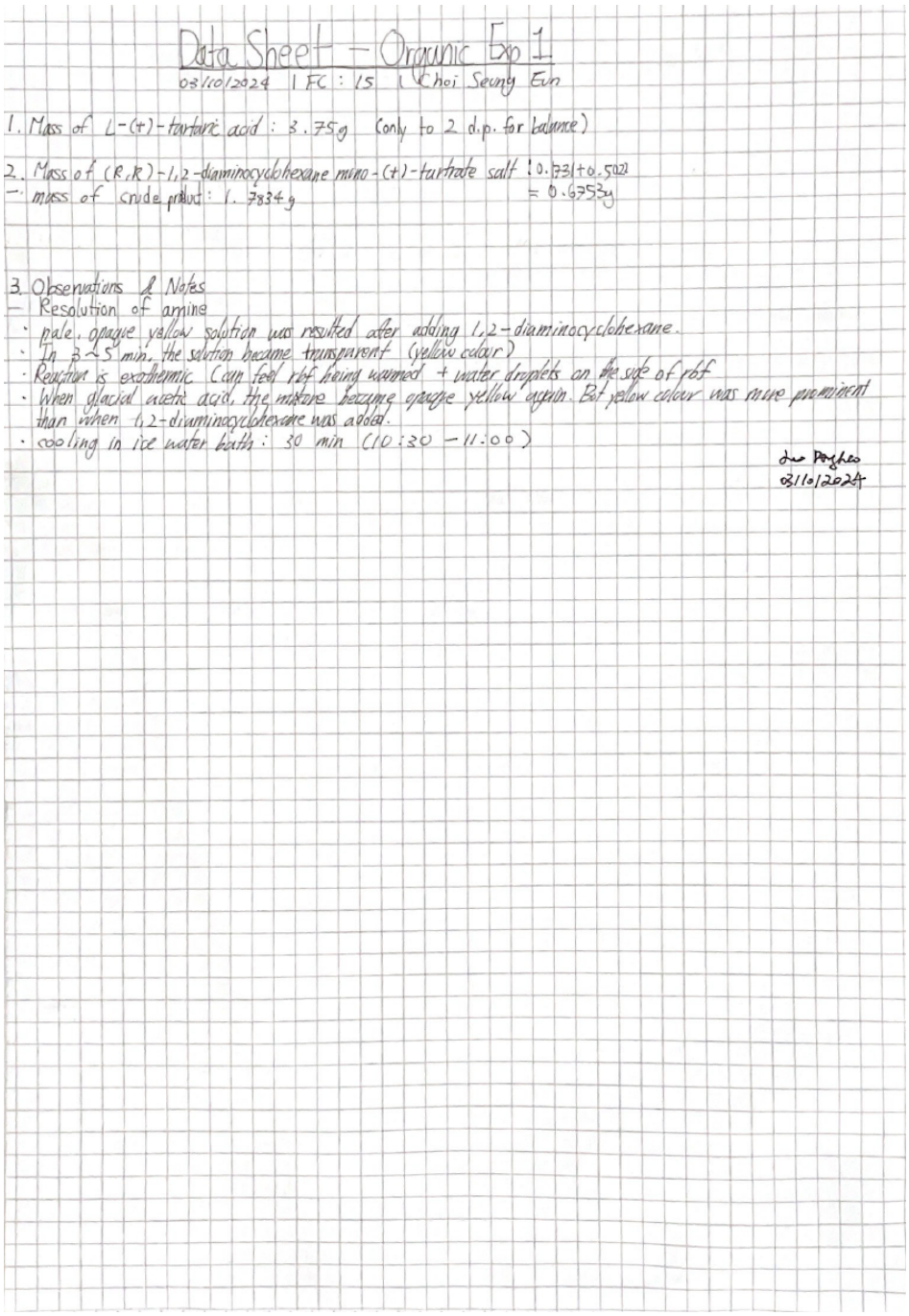
0.5537 (g) ÷ 10 (mL) = 0.05537 g/mL (4 s.f.)

The $[\alpha]_{experimental}$ value at 589 nm: $[\alpha] = \frac{\alpha}{cl} = \frac{-17.40}{0.05537 \times 1.00} = -314.2°$ (4 s.f.)

Since $[\alpha]_{theoretical}$ value is -315° for pure [(*R,R*)-*N,N'*-bis(3,5-di-*tert*-butylsalicylidene)-1,2-cyclohexanediamine],

$$\%\ ee = \frac{\left|[\alpha]_{\text{experimental}}\right|}{\left|[\alpha]_{\text{theoretical}}\right|} \times 100\% = \frac{|-314.2°|}{|-315°|} \times 100\% = 99.75\%\ (4\ s.f.)$$

8.4 Data sheets



Data Sheet – Organic Exp 1
03/10/2024 | FC : 15 | Choi Seung Eun

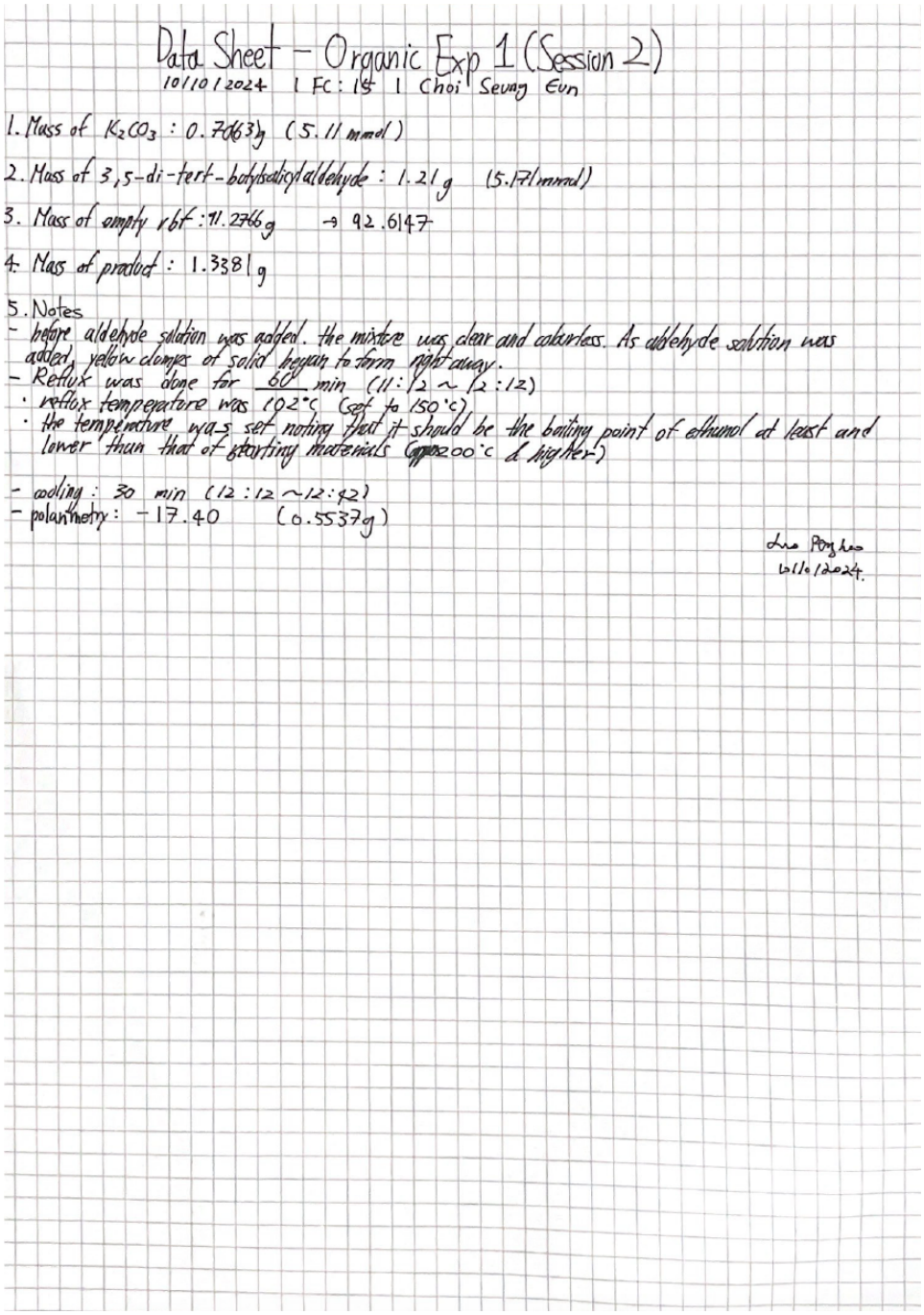
1. Mass of L-(+)-tartaric acid : 3.75g (only to 2 d.p. for balance)

2. Mass of (R,R)-1,2-diaminocyclohexane mono-(+)-tartrate salt (0.731+0.502)
– mass of crude prod : 1.7834g = 0.6753g

3. Observations & Notes
– Resolution of amine
· pale, opaque yellow solution was resulted after adding 1,2-diaminocyclohexane.
· In 3~5 min, the solution became transparent (yellow colour)
· Reaction is exothermic (can feel rbf being warmed + water droplets on the side of rbf
· When glacial acetic acid, the mixture became opaque yellow again. But yellow colour was more prominent than when 1,2-diaminocyclohexane was added.
· cooling in ice water bath : 30 min (10:30 – 11:00)

Lee Peyhee
03/10/2024

Figure A2: Data sheet of session 1 (03/10/2024)



Data Sheet – Organic Exp 1 (Session 2)
10/10/2024 | FC: 15 | Choi Seung Eun

1. Mass of K_2CO_3 : 0.7063g (5.11 mmol)
2. Mass of 3,5-di-tert-butylsalicylaldehyde : 1.21g (5.171mmol)
3. Mass of empty rbf : 91.2746g → 92.6147
4. Mass of product : 1.3381g
5. Notes
- before aldehyde solution was added, the mixture was clear and colorless. As aldehyde solution was added, yellow clumps of solid began to form right away.
- Reflux was done for 60 min (11:12 ~ 12:12)
  - reflux temperature was 102°C (set to 150°C)
  - the temperature was set noting that it should be the boiling point of ethanol at least and lower than that of starting materials (approx 200°C & higher)
- cooling : 30 min (12:12 ~ 12:42)
- polarimetry : −17.40 (0.5537g)

Choi Seung Eun
10/10/2024

Figure A3: Data sheet of session 2 (10/10/2024)